Jason Cabrejos

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Robotics and Embedded Software Engineer

Experience

Burlington Northern Santa Fe (BNSF Railway) *March 2023 – Present*
Robotics Embedded Software Engineer

- Delivered **96%** container location accuracy to within **±1 spot** across **5+ BNSF yards**, saving **$5M per year**
- Shipped a real-time container inventory system from **prototype to production**, designing the architecture and bringing up the **Nvidia Jetson** edge units (embedded Linux, PoE) at **under $10K per unit**
- Engineered the on-device pipeline that reads container IDs (**OCR**) and fuses camera detections with **GPS/RTK** to locate each container's yard spot, feeding a live yard inventory
- Fielded **50 Jetson units** nationwide across multiple hardware stacks and kept the fleet updated with **Docker** and **Azure IoT Hub**
- Built a **Gazebo/Nav2** yard simulator that replaced yard tests with a repeatable baseline and cut validation from **days to hours**
- Led engineers and field installers through fleet rollouts and worked directly with BNSF stakeholders and vendors

Automation and Intelligent Systems Lab, UTA Research Institute *October 2022 – February 2023*
Researcher

- Developed **ROS/OpenCV** object recognition and retrieval scripts for a Baxter robotic arm in community-engagement projects
- Improved sensor stability across multiple platforms, reducing vibration-induced errors in industrial settings

Crown Castle *January 2022 – August 2022*
Robotic Development Intern

- Automated fiber-optic deployment equipment in the field with **ROS, Python**, and Ubuntu Linux, cutting crew size **30%**
- Bridged an industrial vehicle's **CAN bus** to an embedded **Nvidia Jetson** for sensor feedback and actuator control, enabling automated curb-following and cable drift correction

Laboratory of Advanced Materials Manufacturing and Analysis *June 2021 – January 2022*
Researcher

- Designed an Automated Fiber Placement (AFP) tool for a Kuka robotic arm for experimental composite layups
- Wrote the **Python and EtherNet/IP** integration that linked the AFP tool to the Kuka arm and its **PLC** for slicer compatibility
- Oversaw construction of the robotic work cell to meet safety requirements and project specs

Leadership & Awards

SHPE Dallas/Fort Worth *June 2026 – Present*
Professional Development Director, Executive Board Officer

- Own strategy, budget, and delivery for the chapter's professional development pillar across three programs (PDS, TDS, GLEDA)
- Coordinate the PD team with AI-assisted workflows (Notion CRM, automated email marketing, flyer production)

Hackathons
5 podium finishes

- Pitched a camera system to improve child safety without distracting the driver, **1st place**
- Presented a Machine Learning model and business plan to predict potential failures on the railway using big data, **1st place**
- Created a mock-interview agent with Microsoft Copilot Studio and Power Automate that gives practice feedback, **2nd place**
- Trained a TensorFlow computer vision model to automate the recognition and sorting of waste using a robot, **3rd place**
- Prototyped an agricultural IoT system using computer vision, Azure, and Raspberry Pis to time watering and harvest, **3rd place**

Rover Team *August 2017 – August 2021*
President, Arm Leader, Member

- Led a team of 30 at the University Rover Challenge (terrain traversal, object retrieval, life detection), placing **15th of 88 teams**
- Managed 5 people to design and build a 5 DOF robotic arm with Ansys FEA (**2.4 safety factor**) and PID joint control to **±0.4 in**

NASA L'Space Academy *May 2019 – November 2019*
Project Manager

- Led 6 students from across the nation through 2 NASA projects, a submarine concept for Saturn's moon Titan and upgrades to ISS robotic systems, and met project deadlines

Education

University of Texas at Arlington *August 2025*

- Master of Science in Computer Science
  - Focus on Intelligent Systems and Computer Networks
- Bachelor of Science in Mechanical Engineering
  - Minor in Computer Science

Qualifications

- Languages: Python, C++, C#, MATLAB, LabVIEW
- Robotics/ML: ROS/ROS 2, Gazebo, Nav2, OpenCV, TensorFlow, Roboflow
- Tools: Nvidia Jetson, Linux, GPS/RTK, CAN bus, Git, Docker, RabbitMQ, MongoDB, EtherNet/IP, PLCs, Azure, AWS, IoT
- Domains: Robotics, mechatronics, sensor/actuator integration, machine vision, GPS/RTK positioning, simulation
- CAD and Fabrication: Solidworks (CSWA certified, 2016), Solid Edge, 3D printing, CNC, lathe, manual milling